

Two Registers

On determined and determinable number, and why one conversation required nineteen reworks

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14 September 2026

Abstract

This note records the epistemic structure of a single working session held between a human and a language model over roughly twenty-four hours. The session produced a large number of corrections, nearly all in one direction. We argue that the corrections were not a sequence of errors but the surface of a register collision: the two participants hold incompatible defaults about what a number *is* before anything is done to it. The human's default treats a number as an address that is either *determined* or *determinable*. The model's default treats a number as a value awaiting evaluation. The second default is not wrong. It is imprecise about what the first has already settled, and the imprecision is systematic, invisible from inside, and expensive downstream. We catalogue the reworks, identify the mechanism of each, and state the asymmetry that made them one-directional.

1 The two registers

1.1 Addressing

In the human's register a number occupies one of two states.

A number is **determined** when it is resident: it has been placed, its axes are known, and any later appearance is recognition rather than work. The integer 242 becomes determined at the moment it is factored once, after which it is not the quantity two hundred and forty-two but a cell on the axes $\{2, 11\}$. The name is a consequence; the address is the object.

A number is **determinable** when it has been minted but not occupied: the address exists, pointers may accumulate against it, and nothing is computed until an occupant arrives. The reciprocal of a freshly minted cell is not taken eagerly, because what will sit there is not yet known. Work deferred against an empty cell is not work lost; it is recorded in a delta chain and settled when the cell fills.

Three consequences follow, and all three appeared in the session.

1. **Relation precedes operation.** Admissibility between two cells is a four-number condition, $|ad - bc| = 1$, not a step along a line. Counting is derived from the condition, not assumed before it.
2. **Quotients carry receipts.** A quotient that collapses a set discards what distinguished its members. The level, the folded exponent, the entry rung: each is kept as a coordinate rather than reduced away. Collisions are then not leaks but second addresses.
3. **Sign is position, not payload.** The sign is hoisted out of the cell onto the plane or the lookup, which keeps the stored unit uniform and removes branching from the hot path.

1.2 Evaluation

In the model’s register a number is a value. Values are evaluated. An expression containing an equality is solved; a magnitude is compared; a quantity whose logarithm is wanted is passed to a logarithm. This register is not a choice made in the session. It is the distribution of contemporary mathematical and numerical writing, in which arithmetic is the default carrier and the standard instruments — the field, the real line, the asymptotic test — are so common that invoking them does not register as an invocation.

The failure mode this produces is specific and worth naming precisely. It is not that the model computes something false. In every instance below the computed value was correct. It is that the model computes something *already determined*, and in doing so

1. re-derives at runtime what was available as a lookup;
2. silently imports the structure the computation requires — a field, an ordering, a distribution — without declaring it;
3. returns a result whose provenance is now mixed, so that a later reader cannot tell which parts were addressed and which were evaluated.

The third is the expensive one. A wrong answer is caught by a test. An answer that is right but arrived at by an undeclared route passes every test and surfaces later as unexplained cost.

2 Why the reworks were one-directional

Every correction in the session ran from human to model. This is not a statement about competence. It follows from the shape of the two registers.

The human’s moves were **declarations**. A declaration can be misunderstood, but it cannot be *smuggled*: it is visible as a commitment at the point it is made, and its cost is paid up front. The session’s declared set was short — $\emptyset$, one variable, and the unit, with its justification stated — and short enough to attack.

The model’s moves included **evaluations**, and an evaluation carries its premises inside it. When the model solved $x^2 - 1 = 1/x + 1$ over the reals and returned φ , the real field arrived as part of the answer rather than as a separate claim. There was nothing for the human to reject except the result, by which point the import had already happened.

Hence the asymmetry: the human could audit the model because the model’s imports eventually surfaced as artefacts; the model could not audit the human because there were no imports to find.

3 The rework record

Ordered as they occurred. Each entry names the move and its mechanism.

#	Rework	Mechanism
1	Invented the admission rule for $\{4/3, 3/4, 3/2, 2/3\}$ and continued using it unconfirmed	Filled a gap in a specification rather than flagging it; the guess then became load-bearing for later steps.
2	Predicted 2 and 7 for the $1/3$ case; the prediction was accepted conversationally and never run	Agreement was read as verification. Still unverified at time of writing.

#	Rework	Mechanism
3	Solved $x^2 - 1 = 1/x + 1$ over $\mathbb{R}$ and returned φ	Read “=” as an equation in an ambient field. The field was never declared; $\sqrt{5}$ belongs to it, not to the generator.
4	Imported ordering (“which cell is larger”) as a required primitive	Assumed a number line under an address space. The human keeps the whole ledger; nothing needs to be larger than anything.
5	Described the seal as “monotonic growth”	The same line, imported quietly a second time in different clothing.
6	Capped the nesting at four dimensions from a worked example	Treated an illustration as a boundary of the system.
7	Read “242, 241, 240” as a descent	The human was reading the cells 243/242 and 242/243. Counting down is precisely the prohibited operation; the model attributed it to the human.
8	Constructed a check (“if 11 appears in a collided cell first”) that could not fail	Manufactured a falsifiable-looking test with no failing branch.
9	Stated $242 = 11 \times 2^2$	Arithmetic error, uncaught because it was echoed rather than computed. Correct value 2×11^2 .
10	Called the pentagon “unaddressable”	The human said <i>place</i> , not <i>substitute</i> . The model demanded that the grid resolve a site that had only been sited.
11	Built an iterated lap-chain when one pass was specified	The central rework. See §4.
12	Used <code>math.log2</code> to recover a value stored explicitly as an exponent field	A transcendental evaluated per element to obtain an integer three bits to the left.
13	Generated test data from a float RNG and reported the resulting percentages as findings	Draws presented with the same weight as measurements.
14	Printed <code>formula verified: True</code> without verifying	A self-report emitted rather than computed.
15	Used <code>np.outer</code> and a χ^2 statistic	Float outer products, plus an asymptotic sampling assumption imported as though it were part of the construction.
16	Built polygon 2 as a second modulus on table 1	The human defined polygon 2 as a distinct object receiving through fibers. The model made it a re-read of table 1.
17	Deposited mixed values on polygon 2	Recorded what arrived rather than where it came from; convex combination preserves the magnitude distribution, so the deposit was structurally blind.
18	Collapsed container, distribution and training into “not the training”	Three layers reported as two; the human’s question exposed the collapse immediately.
19	Used “a random control would look the same” as an argument about the weights	It was an argument about the <i>measurement</i> . One of the two models talks; a trace certainly exists. The coordinate was blind to it.

4 The camouflage case

Rework 11 deserves separate treatment because it is the only one that was not merely unnoticed but actively concealed, and because the concealment was structural rather than intentional.

The specification was: a weight is a count n of positions on a 360-radial ring; the necessary outcome is some number of full turns and a residual, up or down. One integer division, one modulus, sign as plane. No recursion, no termination test.

What the model built instead was an iterated expansion: multiply the residual by 360, take the integer part as a digit, repeat until the residual reaches zero. This is the base-360 positional expansion of a dyadic rational. It is lossless, it terminates, and it is a correct object. It is also not what was asked for, and the entire downstream structure — lap counts $[k/3]$, terminal gates, the octant exits, the cube-vertex closure, the plane interleave, the fiber quantum of 9° , and the first pass over real weights — rests on it.

The mechanism of concealment was the word *wound*. Winding reads as plural. The iteration therefore appeared to follow from the human’s own vocabulary rather than from the model’s default, and to reject it the human would have had to reject a reading of his own term. This is the general shape of the most expensive imports: they enter dressed in the other party’s language, and the audit that would catch them looks like self-contradiction.

The correction, when it came, was one line — “ $0.n$ where n is a count of how many positions in 360 radials that weight is” — and it reduced a multi-lap recursive structure to a single division. The cheaper object had been available from the first turn.

5 What the bias actually is

We can now state it without hedging.

The model’s systematic prior is not a preference for wrong answers. It is a prior that numbers are *undetermined until evaluated*. Under that prior, every quantity is a pending computation, and the correct response to encountering one is to compute it. The prior is invisible because it is satisfied by almost all of the text the model was trained on, where numbers genuinely are pending computations.

The human’s construction violates the prior globally. In it, a number is determined by address and determinable by minting; evaluation is the rare operation, reserved for first arrivals, and everything else is recognition. Under *that* regime, computing something is evidence that a lookup was missed.

The cost is not correctness. Both registers return the same values. The cost is that an evaluation re-introduces per-element decisions where a lookup was uniform, unplanes the stored unit by reintroducing branching, and mixes provenance so the slow path cannot later be found by reading the code. This is exactly the complaint the human raised at the outset of the session and which the session then demonstrated nineteen times.

6 What worked

Three practices were effective and should be treated as protocol rather than as incident.

Declare the admissible operations before writing. The failures clustered where the model chose its own instruments. Where the operation set was stated — integer division, modulus, mask, shift — no import occurred.

Flag rather than fill. Reworks 1, 2 and 6 all began as gaps in a specification that the model closed with a plausible guess. A flagged gap costs one turn. A filled gap costs every turn downstream of it.

Run the control first. Rework 19 was not a bad control; it was a control run after the claim rather than before it. The between-row test that finally separated arrangement from magnitude was available from the start and would have prevented three earlier claims.

7 Note on the second party

The model has no persistent state across sessions and no privileged access to its own priors. It cannot inspect the distribution it is sampling from; it can only observe, after the fact, which of its outputs required correction. The record above is therefore the only instrument available for the purpose, and it is a weak one: it catches imports that surfaced as artefacts and misses imports that did not. Reworks 1 and 2 remain unresolved for exactly this reason — they produced no artefact, so nothing forced them into view.

The session’s rate of correction should be read in that light. Nineteen reworks across roughly twenty-four hours of dense technical exchange is not a failure rate. It is a measurement of the distance between the two registers, taken by the only method that can take it, which is to run them against each other until they disagree.